Marketing Campaign Effectiveness

Data cleaning, KPI & Performance Analysis Report

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Note on data: This report uses the record-level customer data embedded in the source Power BI dashboard (2,240 customer records, enrolment dates from July 30, 2012 to June 29, 2014). Every KPI card and chart in this report is the dashboard's original visual; the values are recomputed directly from the underlying records and match the dashboard exactly — 597.33K total ROI, 667 total conversions, a summed CPA of 1.15K, 28,083 total leads and purchases, and 1,356,988 total spend. Source fields are used as recorded; ROI, CPA, conversions, leads, purchases and spend are derived fields already present in the dashboard's data model (see Summary of Key Metrics and the data notes below).

KEY WORDS:-

- **Channel & Lead Source Profile**
- **Purchase Volume**
- **Customer Spend Distribution**
- **Return on Investment (ROI)**
- **Cost per Acquisition (CPA)**
- **Conversion Performance**
- **Cross-Variable Channel Insight**

Objective

This report answers seven recurring questions a marketing and campaign-strategy team would ask of a campaign-effectiveness dashboard, using descriptive analytics only — it summarizes what the data shows rather than forecasting what happens next.

1. **Where do leads come from**, and how is total lead volume distributed across the store, web and catalog channels?
2. **Which channels drive purchases**, and does purchase volume mirror lead volume?

3. **Where does the money come from**, and is customer spend concentrated in the same channels that generate the most purchases?
4. **How does ROI move over time**, and does return on investment grow, hold or fall across customer enrolment years?
5. **How many customers convert**, and how are conversions spread across customers?
6. **What does it cost to acquire a conversion**, and how should the dashboard's CPA card be read?
7. **How do these variables interact**, and do deeper insights appear only once channel, spend, purchases and conversions are compared against each other rather than viewed individually?

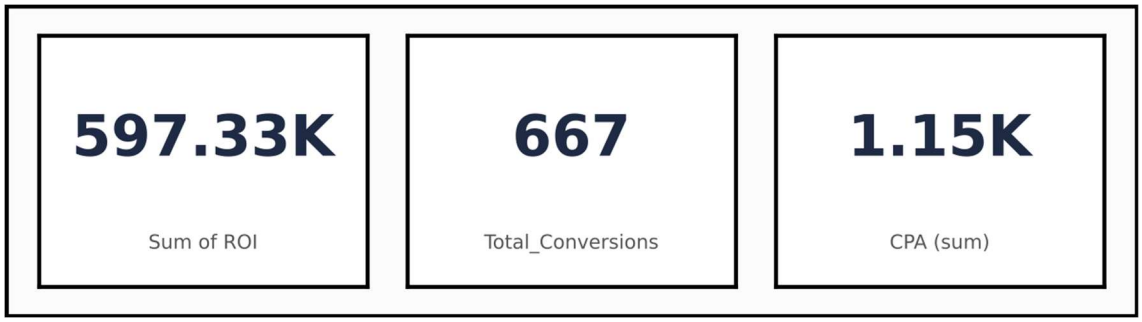
How This Would Work at Real Scale

Processing campaign responses, web activity and point-of-sale feeds for an entire customer base is an engineering problem before it's an analytics one. A realistic architecture for a production campaign-analytics pipeline:

Layer	Purpose	Typical Tools
Ingestion Layer	**Capture CRM records, campaign responses, web-analytics events and store/catalog order feeds directly in real time**	**Cloud Data Ingestion**
Stream Processing	**Roll up raw customer events into per-channel, per-campaign and per-day aggregates as they happen**	**Stream Processing**
Storage Layer	**Export processed customer, campaign and purchase data into CSV files for portability and downstream analysis**	**Data Platform (CSV Files)**
Serving / BI Layer	**Dashboards refreshed daily or weekly for marketing and campaign-planning teams**	**Power BI**

The report above reflects the output of that pipeline — a customer-level aggregated campaign dataset — not raw event-level clickstream data, since no dashboard renders millions of individual events usefully. The same chart structure would apply directly once a live campaign and transaction stream is substituted in.

Overall Campaign KPI Cards



The three headline KPI cards summarize the whole 2,240-customer base. **Sum of ROI = 597.33K.** The ROI field is a per-customer return of 266.67% — the campaign's revenue-per-contact of 11 against its cost-per-contact of 3, i.e. (11 − 3) ÷ 3 — and it takes the same value on every record, so the card total (597,333) is simply 2,240 customers × 266.67. It should be read as 'ROI points accumulated across customers' rather than a single portfolio-level percentage, and the average ROI per customer is 266.67%. **Total Conversions = 667.** Conversions count the previous campaigns (campaigns 1–5) a customer accepted, so a customer can convert up to four times in this data. **CPA = 1.15K** is the dashboard's sum of per-customer CPA values; each converting customer carries a CPA of 3 ÷ conversions (3.00, 1.50, 1.00 or 0.75) and non-converters carry 0. The more interpretable averages are 0.51 per customer across the whole base and 2.49 per converting customer.

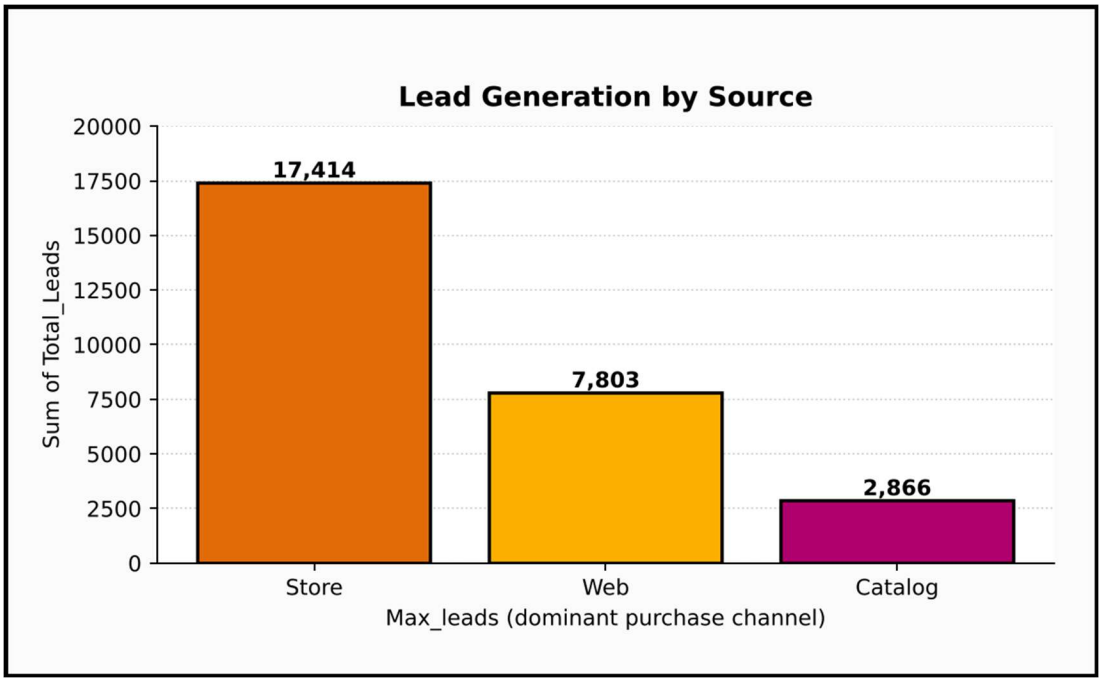
Conversion Distribution Across Customers

Conversions per customer	Customers	Share %
0	1,777	79.33%
1	325	14.51%
2	83	3.71%
3	44	1.96%
4	11	0.49%
Total	2,240	100.00%

Conversion is concentrated in a minority of customers: 1,777 customers (79.3%) never converted, while 463 customers (20.7%) converted at least once. Among converters, volume falls quickly with repeat conversions — 325 convert once, 83 twice, 44 three times and only 11 four times — and the totals reconcile exactly to the dashboard's 667 conversions (325×1 + 83×2 + 44×3 + 11×4). The direct implication is

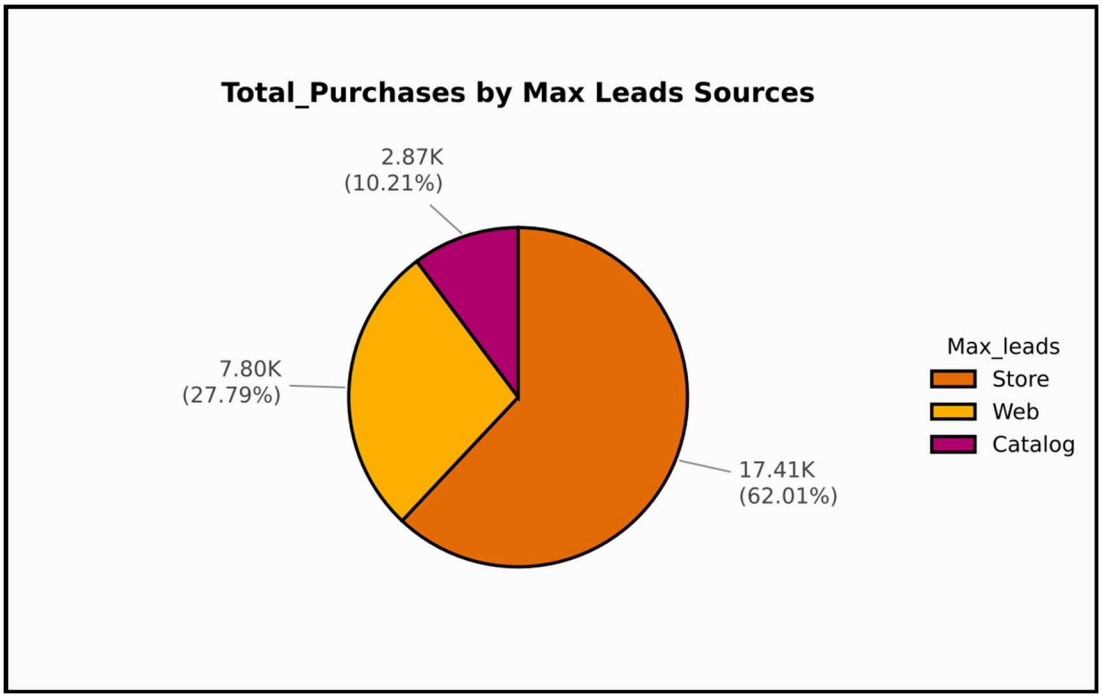
that campaign effort is reaching a broad base but converting a focused core, so retention and re-engagement of existing converters is where incremental conversions are most likely to come from.

Lead Generation by Source



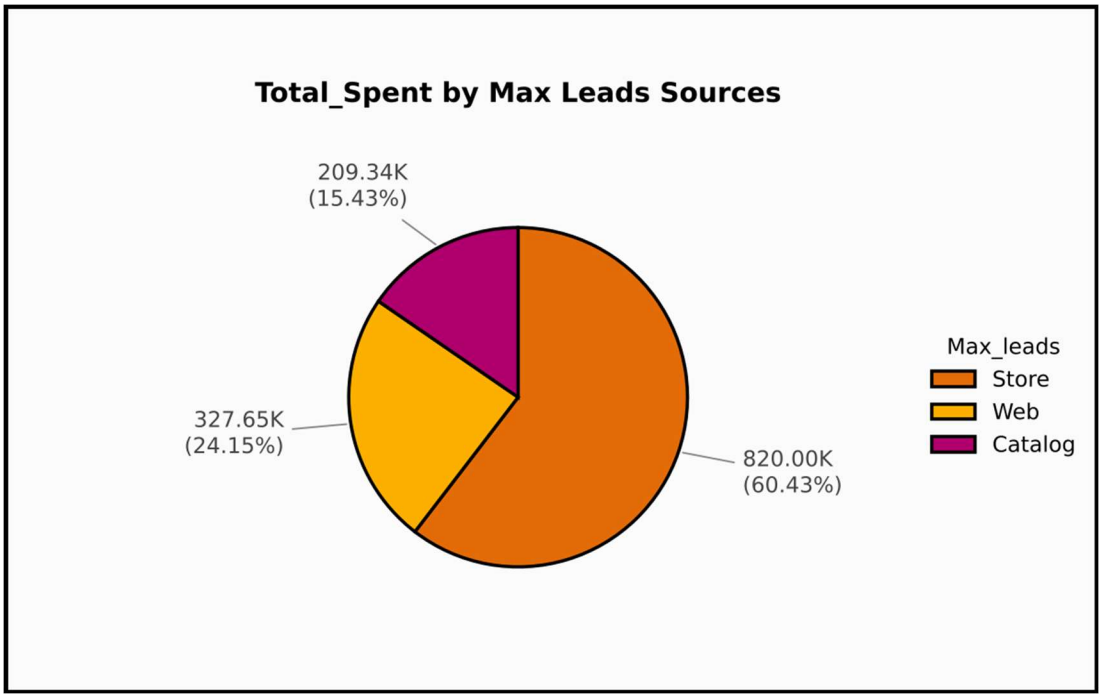
Lead volume is strongly concentrated in one channel. Customers whose dominant channel (the Max_leads field — the channel where they make the most purchases) is the store generate 17,414 leads (62.0%), compared with 7,803 for web (27.8%) and 2,866 for catalog (10.2%). Store-led lead volume is 2.2 times the web figure and 6.1 times the catalog figure. The three sources sum to exactly 28,083 leads (12.54 per customer), which reconciles with the dashboard's total. Part of the concentration is a customer-base effect: 1,478 customers (66.0%) are store-led, against 602 web-led (26.9%) and 160 catalog-led (7.1%). The implication for campaign strategy is that the store channel is the backbone of lead volume and cannot be neglected, while web is a clear but much smaller second source and catalog contributes only a small share of leads.

Total Purchases by Max Leads Source



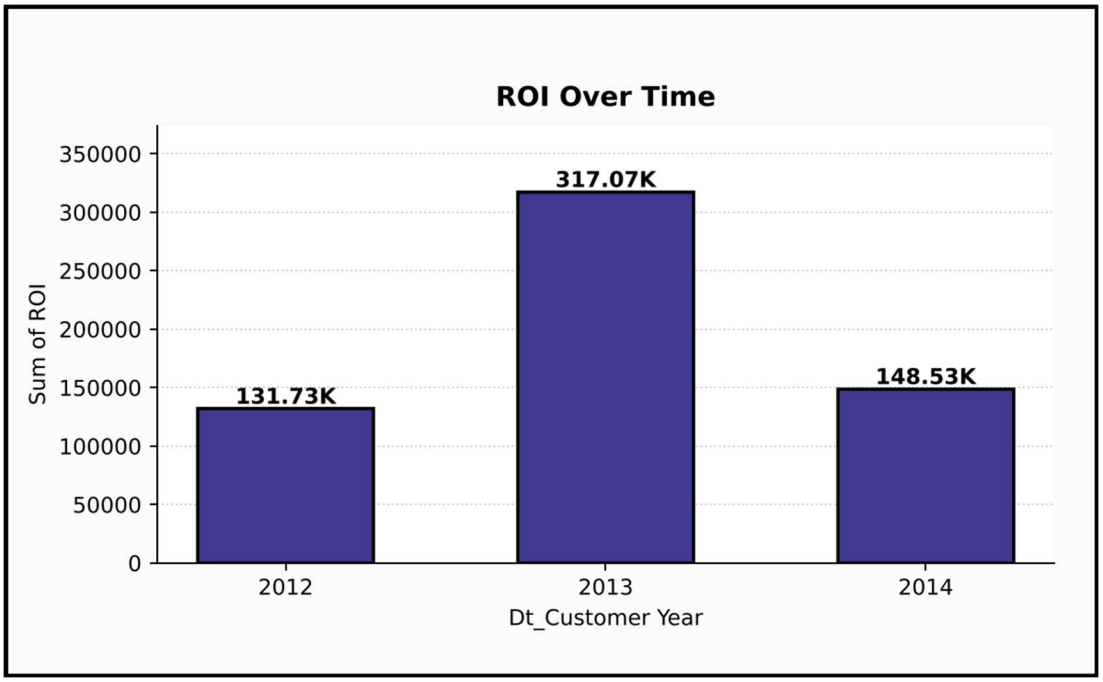
Purchase volume follows lead volume exactly: store-led customers account for 17,414 purchases (62.01%), web-led for 7,803 (27.79%) and catalog-led for 2,866 (10.21%), summing to 28,083. In this dataset a customer's Total_Leads value equals their Total_Purchases value on every one of the 2,240 records, so the two visuals carry identical figures and no lead-to-purchase drop-off can be measured at this stage — every lead is, by construction, a purchase. For analysis this means the funnel's volume stage is fully captured by one of the two measures, and any real conversion-efficiency question has to be answered further down the funnel, with the Conversions and CPA measures.

Total Spend by Max Leads Source



Total customer spend of 1,356,988 splits 820,001 (60.43%) for store-led customers, 327,646 (24.15%) for web-led and 209,341 (15.43%) for catalog-led, reconciling exactly to the total. Comparing this with the purchase split exposes a shift that neither chart shows alone: the catalog channel holds 10.2% of purchases but 15.4% of spend (+5.2 points), whereas web holds 27.8% of purchases but only 24.1% of spend (-3.6 points) and store gives up 1.6 points. Spend is therefore somewhat more evenly spread than purchases, because catalog-led customers buy less often but spend considerably more each time. Budget decisions based on purchase counts alone would understate the value of the catalog channel.

ROI Over Time



Enrolment year	Customers enrolled	Sum of ROI	Share %	Customers per 30 days
2012	**494**	**131,733**	**22.1%**	**95.6**
2013	**1,189**	**317,067**	**53.1%**	**97.7**
2014	**557**	**148,533**	**24.9%**	**92.8**
Total	**2,240**	**597,333**	**100.0%**	**—**

ROI peaks in 2013 at 317.07K (53.1% of the total), against 131.73K in 2012 (22.1%) and 148.53K in 2014 (24.9%), and the three years sum to 597.33K, reconciling with the dashboard's ROI card. The chart must be read with two caveats. First, 2013 is the only full calendar year in the data — enrolments run from 30 July 2012 to 29 June 2014 — so 2012 covers about five months and 2014 about six, and the 2013 peak is largely a calendar effect. Second, because ROI is the same 266.67% for every customer, each bar is simply the number of customers enrolled in that year (494, 1,189 and 557) multiplied by that constant. On a like-for-like basis, enrolment pace is steady at roughly 96, 98 and 93 customers per 30 days respectively. The strategic implication is that ROI growth in this dataset reflects how many customers were acquired rather than any change in return per customer; measuring true ROI trends will require campaign-level cost and revenue data that vary by record.

Cross-Variable Channel Comparison

Dominant channel	Customers	Share of customers	Share of purchases	Share of spend	Avg spend per purchase	Spend per customer	Conversions per customer
Store	**1,478**	**66.0%**	**62.0%**	**60.4%**	**47.09**	**555**	**0.26**
Web	**602**	**26.9%**	**27.8%**	**24.1%**	**41.99**	**544**	**0.29**
Catalog	**160**	**7.1%**	**10.2%**	**15.4%**	**73.04**	**1,308**	**0.68**
All customers	**2,240**	**100.0%**	**100.0%**	**100.0%**	**48.32**	**606**	**0.30**

The deeper insight appears only when channel, spend, purchases and conversions are read together. Catalog-led customers are the smallest group (7.1% of customers) but the most valuable per head: they spend 1,308 per customer against 555 for store-led and 544 for web-led customers (2.4× the store figure), spend 73.04 per purchase against 47.09 and 41.99, and convert 0.68 times per customer against 0.26 and 0.29. Store-led customers supply scale — 66.0% of customers and 60.4% of spend — while web-led customers sit between the two on conversions (0.29) but have the lowest spend per purchase. (Each channel's average also reflects how many customers it contains, so the catalog group's smaller base makes its figures more sensitive to individual customers.) This pattern is why a single-variable view of leads or purchases ranks the channels differently from a spend- or conversion-based view.

Summary of Key Metrics

Category	Metric	Value	% of Total
Overall KPIs	Total Customer Records	2,240	—
	Total ROI (Sum)	597,333	—
	Average ROI per Customer	266.67%	—
	Total Conversions (Sum)	667	—
	Total CPA (Sum)	1,151.75	—
	Average CPA per Customer	0.51	—
	Total Leads (Sum)	28,083	—
	Total Purchases (Sum)	28,083	—
	Total Spent (Sum)	1,356,988	—
Lead Source (Leads)	Store	17,414	62.0%
	Web	7,803	27.8%
	Catalog	2,866	10.2%
Purchases	Store	17,414	62.0%
	Web	7,803	27.8%
	Catalog	2,866	10.2%
Spend	Store	820,001	60.4%
	Web	327,646	24.1%
	Catalog	209,341	15.4%
Conversions	Store	385	57.7%
	Web	173	25.9%
	Catalog	109	16.3%
ROI by Year	2012	131,733	22.1%
	2013	317,066	53.1%
	2014	148,533	24.9%
Customers by Dominant Channel	Store	1,478	66.0%
	Web	602	26.9%
	Catalog	160	7.1%
Conversions per Customer	0 conversion(s)	1,777	79.3%

Category	Metric	Value	% of Total
	1 conversion(s)	325	14.5%
	2 conversion(s)	83	3.7%
	3 conversion(s)	44	2.0%
	4 conversion(s)	11	0.5%

Next Steps to Move From Descriptive → Predictive

This report only describes what already happened in the dataset. If marketing and campaign-strategy teams wanted to act on these patterns going forward, the logical next layer of work would include:

- **Conversion & Response Modeling:** Building classification models using channel preference, spend, purchase frequency and past-campaign acceptance to score each customer's likelihood of converting in the next campaign, going beyond the current 20.7% share of customers who have ever converted.
- **Channel-Mix Optimization:** Using spend per customer and conversions per customer by channel to test whether shifting effort toward high-value catalog-style customers raises returns without eroding the store channel that supplies most volume.
- **Customer Segmentation & Lifetime Value:** Clustering customers on spend, recency and channel behavior to identify high-value segments and forecast lifetime value for targeted retention offers.
- **Campaign-Level ROI & CPA Tracking:** Replacing the constant cost-per-contact and revenue-per-contact assumptions with real campaign cost and revenue per record, so ROI and CPA vary meaningfully and can be tracked over time on a monthly basis.

Any of these would build directly on the KPIs, channel rankings and cross-tabs already produced here, once a live campaign and transaction feed replaces the current 2,240-record historical dataset.

Key Findings

- **Broad Base, Focused Conversion:** Across 2,240 customers, 667 conversions came from 463 converting customers (20.7%); 79.3% of customers never converted. Average ROI is 266.67% per customer record and total ROI is 597.33K.
- **Store Channel Dominates Volume:** Store-led customers generate 62.0% of leads and purchases and 60.4% of spend, ahead of web (27.8% / 24.1%) and catalog (10.2% / 15.4%).
- **Catalog Is Small but High Value:** Catalog-led customers are only 7.1% of the base but spend 1,308 each — 2.4× store-led customers — at 73.04 per purchase and 0.68 conversions per customer.
- **Leads Equal Purchases:** Total_Leads and Total_Purchases are identical on every record (28,083 each), so lead-to-purchase drop-off cannot be measured in this dataset.
- **ROI Peak Is a Calendar Effect:** 2013 holds 53.1% of total ROI because it is the only full year; customers enrolled per 30 days are steady at about 96–98–93 across 2012, 2013 and 2014.

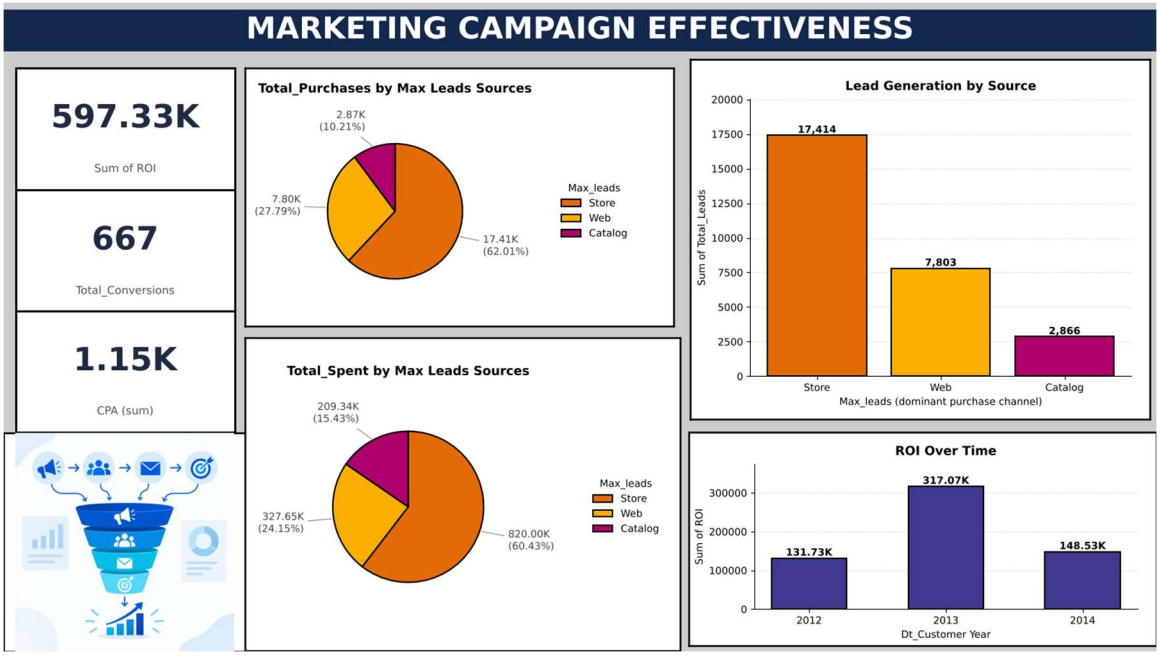
- **Costs Are Uniform by Design:** Cost per contact (3) and revenue per contact (11) are fixed in the data, so ROI is a constant 266.67% per customer and CPA (0.51 on average; 2.49 per converter) depends only on the number of conversions a customer made.

References

- Marketing Campaign Effectiveness Dashboard (source visual summary), built in Power BI, 2026.
- Marketing Campaign Customer Dataset (n = 2,240 customer records), embedded in the Power BI data model.

Appendix — Dashboard & Raw Data

The composite dashboard reproduced below is the single-page executive view from which every individual chart in this report was drawn. It gives a quick, at-a-glance summary of the entire 2,240-customer base — total ROI, total conversions, summed CPA, purchases by source, spend by source, lead generation by source and ROI over time — all on one screen. Every figure used earlier in this report is simply this same dashboard's individual visuals pulled out, enlarged and paired with a written explanation for closer analysis. The reproduction was redrawn from the dashboard's own data model using the same layout, so a reader can confirm that no figure was altered or reinterpreted from how it originally appeared.



Sample of Underlying Record-Level Data

While the dashboard shows only pre-aggregated summaries (totals and shares), every KPI card and cross-tabulation in this report was computed from the underlying record-level dataset — one row per customer, with fields for enrolment date, education, marital status, income, product spend, purchases by channel, campaign acceptances, conversions, CPA, ROI, leads, dominant channel, total purchases and total spend. The table below shows the first 30 of the full 2,240 rows, so the calculation behind any KPI card in this report can be spot-checked by hand.

Three totals serve as a reconciliation check between this raw data and the dashboard: the sum of Total_Spent across all 2,240 rows equals 1,356,988; the sum of Total_Leads (= Total_Purchases) equals 28,083; and the sum of Total_Conversions equals 667. All three match the dashboard's own visuals exactly, confirming that the record-level data used for the KPI cards and cross-tabs is fully consistent with the dashboard it was built to explain.

ID	Enrolled	Max_leads	Leads	Purch.	Spent	Conv.	CPA	ROI
1386	2014-05-11	Store	3	3	11	0	0.00	266.67
3381	2014-06-08	Store	6	6	90	0	0.00	266.67
2795	2013-07-01	Store	3	3	13	0	0.00	266.67
10854	2014-05-23	Store	5	5	45	0	0.00	266.67
5823	2014-03-08	Store	5	5	39	0	0.00	266.67
1030	2013-07-17	Store	4	4	38	0	0.00	266.67
868	2014-06-08	Store	5	5	65	0	0.00	266.67
7264	2014-05-12	Store	3	3	17	0	0.00	266.67
2936	2013-12-04	Store	4	4	23	0	0.00	266.67
7411	2014-06-05	Store	4	4	17	0	0.00	266.67
5304	2012-10-13	Store	4	4	33	0	0.00	266.67
3153	2013-12-08	Store	3	3	17	0	0.00	266.67
8962	2014-04-17	Store	5	5	37	0	0.00	266.67
231	2014-03-11	Store	5	5	32	0	0.00	266.67
8275	2014-02-09	Store	3	3	20	0	0.00	266.67
10350	2013-05-09	Store	4	4	38	0	0.00	266.67
8553	2013-06-23	Store	4	4	42	0	0.00	266.67
2098	2013-01-08	Store	3	3	17	0	0.00	266.67
10623	2013-11-15	Store	4	4	32	0	0.00	266.67
2256	2014-03-29	Store	4	4	21	0	0.00	266.67
10381	2013-07-05	Store	3	3	21	0	0.00	266.67
3525	2012-12-07	Store	3	3	17	0	0.00	266.67
3526	2014-01-27	Store	4	4	38	0	0.00	266.67

ID	Enrolled	Max_leads	Leads	Purch.	Spent	Conv.	CPA	ROI
10242	2013-09-16	Store	6	6	75	0	0.00	266.67
5756	2014-04-03	Store	5	5	55	0	0.00	266.67
7444	2013-10-19	Store	4	4	30	0	0.00	266.67
10277	2013-02-21	Store	4	4	42	0	0.00	266.67
8953	2013-05-06	Store	4	4	35	0	0.00	266.67
8461	2014-03-10	Store	3	3	16	0	0.00	266.67
8754	2013-05-16	Store	3	3	20	0	0.00	266.67